

ICE 3107: Digital Signal Processing

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Lecture Outcomes

Compute the Z-transform and inverse Z-transform of discrete-time signals using standard formulas and properties.



Z-Transform

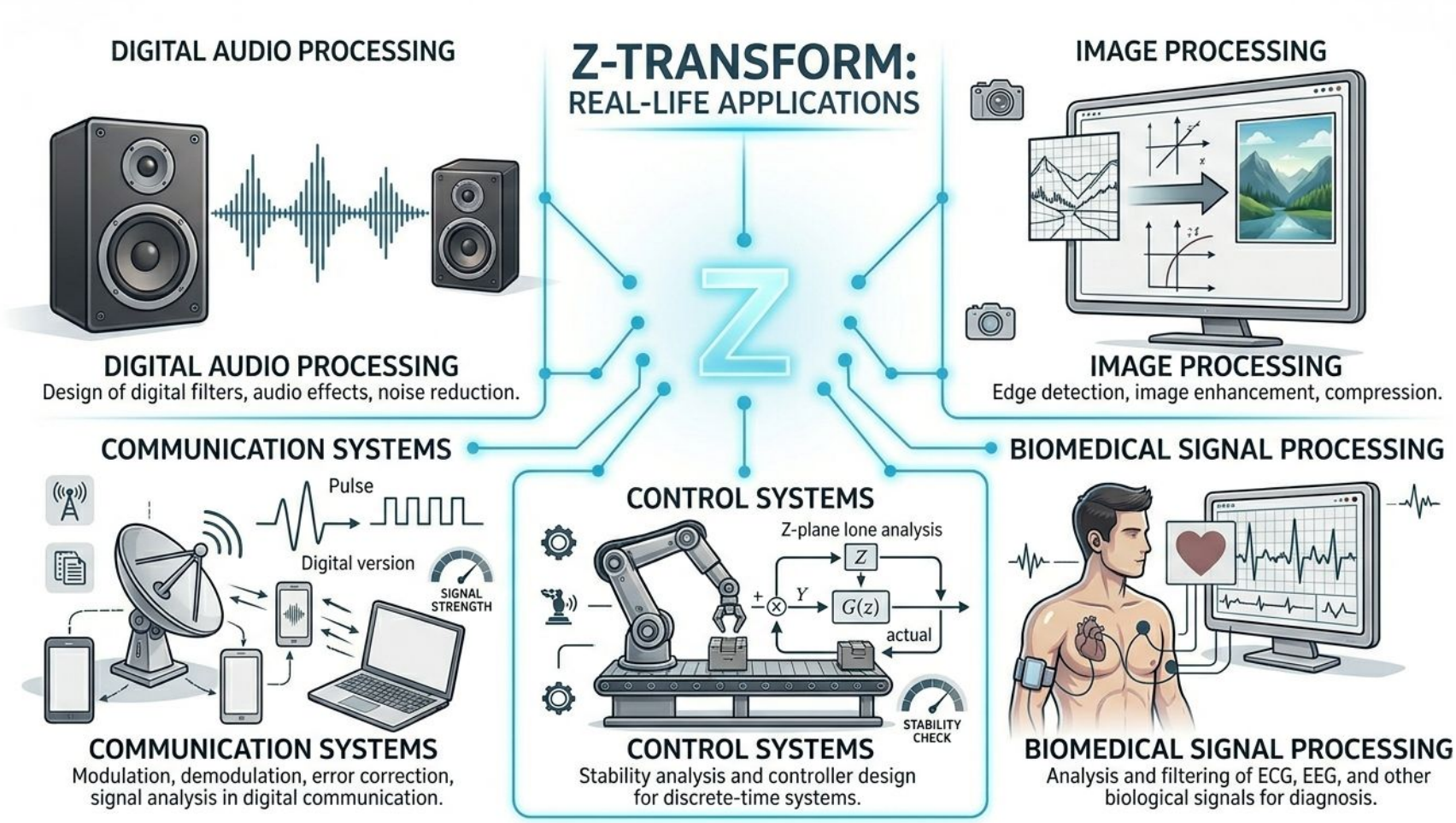
The Z-Transform is the discrete-time equivalent of the Laplace Transform. It converts a discrete-time signal x[n] from the time domain into a complex frequency domain representation X(z), where z is a complex variable.

Z-Transform (Cont.)

Why to use Z-Transform?

- Converts discrete-time difference equations into simple algebraic equations
- Turns convolution in time domain into multiplication in Z-domain
- Helps analyze system properties like stability and causality using poles and ROC

Z-Transform Applications



Z-Transform (Cont.)

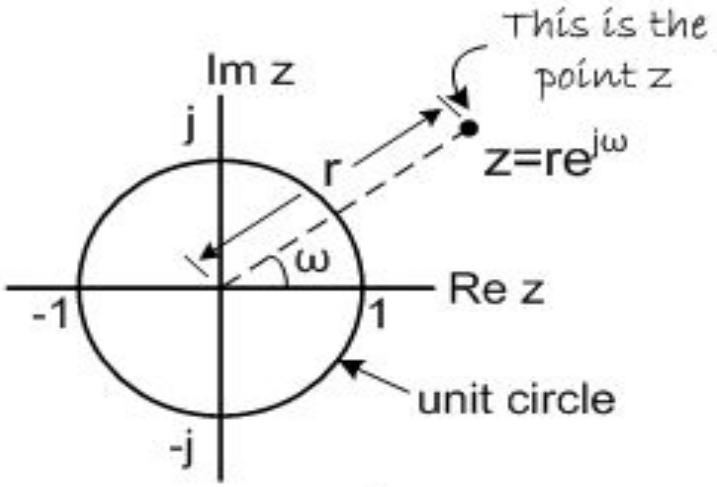
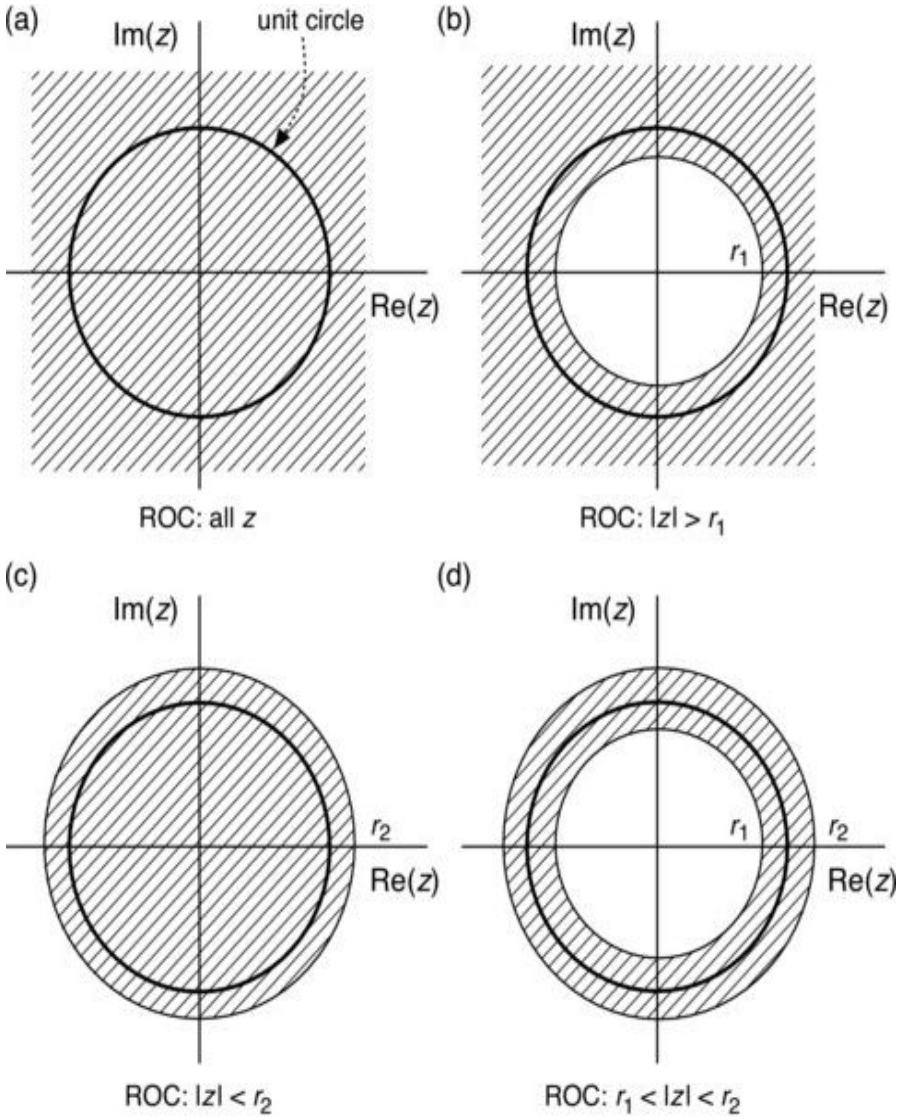


Fig. The point $z=re^{j\omega}$ in the complex z-plane

- z is a complex variable that takes on continuous values.
- $z=\mathrm{Re}(z) + j\,\mathrm{Im}(z) = re^{j\omega}$; where, $r=|z|$, and ω is the angle of z.
- The contour $|z|=1$ is a circle in the z-plane of unity radius. The circle is called the Unit Circle (UC).
- For, $r=1$ ($|z|=1$), the z-transform X(z) of x(n) reduces to its Fourier Transform $F(e^{jw})$.

Region of Convergence (ROC)

- Region of Convergence is the range of complex variable Z in the Z-plane. The Z-transformation of the signal is finite or convergent.
- So, ROC represents those set of values of Z, for which X(Z) has a finite value.



Z-Transform (Cont.)

Example 3.1.1(a):

$$x(n) = \{\underset{\uparrow}{1},2,5,7,0,1\}$$

$$\text{We know,} \quad X(z) = \sum_{n=-\infty}^{\infty} x(n) z^{-n}$$

$$= \sum_{n=0}^{5} \{1,2,5,7,01\} z^{-n}$$

$$= 1z^{-0} + 2z^{-1} + 5z^{-2} + 7z^{-3} + 0z^{-4} + 1z^{-5}$$

$$= 1 + 2z^{-1} + 5z^{-2} + 7z^{-3} + 1z^{-5}$$

ROC: ROC is everywhere except at z=0, because X(z) is infinite for z=0.

Z-Transform (Cont.)

Example 02:

$$x(n) = \{1,2,\underset{\uparrow}{3},4,0,5\}$$

$$\textit{We know,} \quad X(z) = \sum_{n=-2}^{3} x(n) z^{-n}$$

$$= \sum_{n=-2}^{3} \{1,2,3,4,0,5\} z^{-n}$$

$$= 1z^{+2} + 2z^{+1} + 3z^{0} + 4z^{-1} + 0z^{-2} + 5z^{-3}$$

$$= z^{2} + 2z^{1} + 3 + 4z^{-1} + 5z^{-3}$$

ROC: Here, X(z) is infinite if z=0 or if z=infinity. Thus, ROC is everywhere except at z=0 and z= infinity

Z-Transform (Cont.)

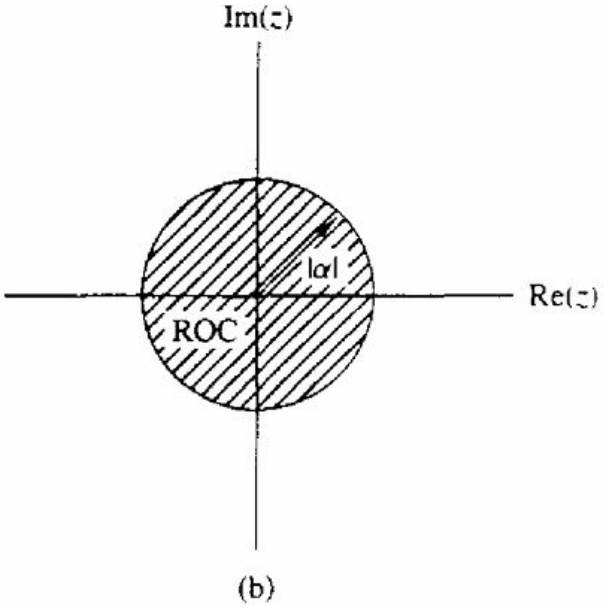
Example 3.1.4

Determine the z-transform of the signal:

$$x(n) = -a^n u(-n-1) = \begin{cases} 0, & n \geq 0 \\ -a^n, & n \leq -1 \end{cases}$$

Ans:

$$X(z) = \frac{z}{z-a}, \quad \text{ROC: } |z| < |a|$$



Z-Transform (Cont.)

Example 3.1.5

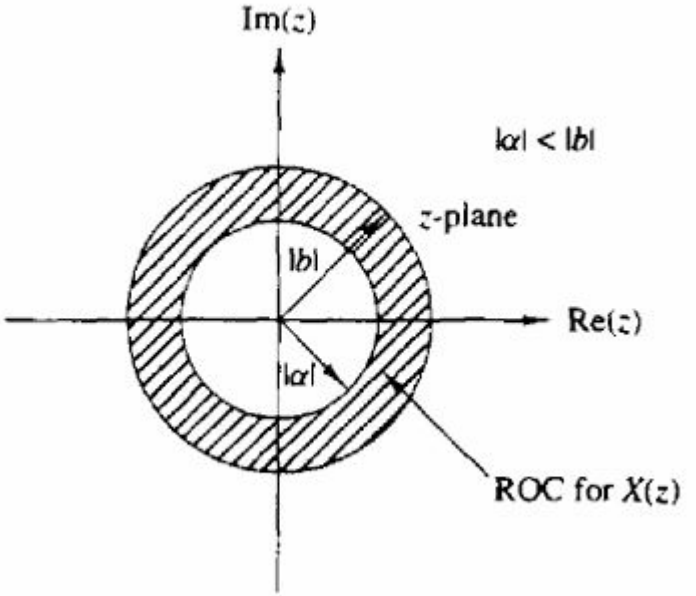
Determine the z-transform of the signal:

$$x(n) = a^n u(n) + b^n u(-n-1)$$

Ans:

$$X(z) = \frac{z}{z-a} - \frac{z}{z-b}$$

$$|a| < |z| < |b|$$



Inverse Z-Transform

The inverse Z-transform recovers the discrete-time signal x[n] from its Z-domain representation X(z). It is the reverse operation of the Z-transform.

Core Formula:

$$x[n] = \frac{1}{2\pi j} \oint X(z) z^{n-1} dz$$

Inverse Z-Transform (Cont.)

Problem: Convert the signal into discrete time signal and find the sequence of x(n):

$$X(z) = 3z^2 + 4 - 2z^{-1} + 3z^{-3}$$

Ans:

$$x[n] = 3\delta[n+2] + 4\delta[n] - 2\delta[n-1] + 3\delta[n-3]$$

$$x[n] = \{3, 0, \uparrow 4, -2, 0, 3\}$$

Practice: Convert the signal into discrete time signal and find the sequence of x(n):

$$X(z) = z^2 + 2z^1 + 3 + 4z^{-1} + 5z^{-3}$$

Properties of the Z-Transform

Linearity:

If

$$x_1(n) \overset{z}{\longleftrightarrow} X_1(z)$$

and

$$x_2(n) \overset{z}{\longleftrightarrow} X_2(z)$$

then

$$x(n) = a_1 x_1(n) + a_2 x_2(n) \overset{z}{\longleftrightarrow} X(z) = a_1 X_1(z) + a_2 X_2(z)$$

Properties of the Z-Transform (Cont.)

Example 3.2.1:

Determine the z-transform and the ROC of the signal:

$$x(n) = [3(2^n) - 4(3^n)]\, u(n)$$

Ans:

$$X(z) = \frac{3}{1 - 2z^{-1}} - \frac{4}{1 - 3z^{-1}} \qquad \text{ROC: } |z| > 3$$

Properties of the Z-Transform (Cont.)

Time shifting:

If

$$x(n) \overset{Z}{\longleftrightarrow} X(z)$$

then

$$x(n-k) \overset{Z}{\longleftrightarrow} z^{-k}X(z)$$

Properties of the Z-Transform (Cont.)

Problem:

Using the time shifting property, find the z-transform of $x_2(n)$ and $x_3(n)$ using $x_1(n)$:

$x_1(n)=[1, 2, 5, 7, 0, 1]$

$x_2(n)=[1, 2, 5\uparrow, 7, 0, 1]$

$x_3(n)=[0, 0, 1, 2, 5, 7, 0, 1]$

Ans:

(a) $X_1(z) = 1 + 2z^{-1} + 5z^{-2} + 7z^{-3} + z^{-5}$, ROC: entire z-plane except $z = 0$

(b) $X_2(z) = z^2 + 2z + 5 + 7z^{-1} + z^{-3}$, ROC: entire z-plane except $z = 0$ and $z = \infty$

(c) $X_3(z) = z^{-2} + 2z^{-3} + 5z^{-4} + 7z^{-5} + z^{-7}$, ROC: entire z-plane except $z = 0$

Thank You


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